

Project Workflow: Week 2

Phase: Software Design

Prepared By:

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1. Objective

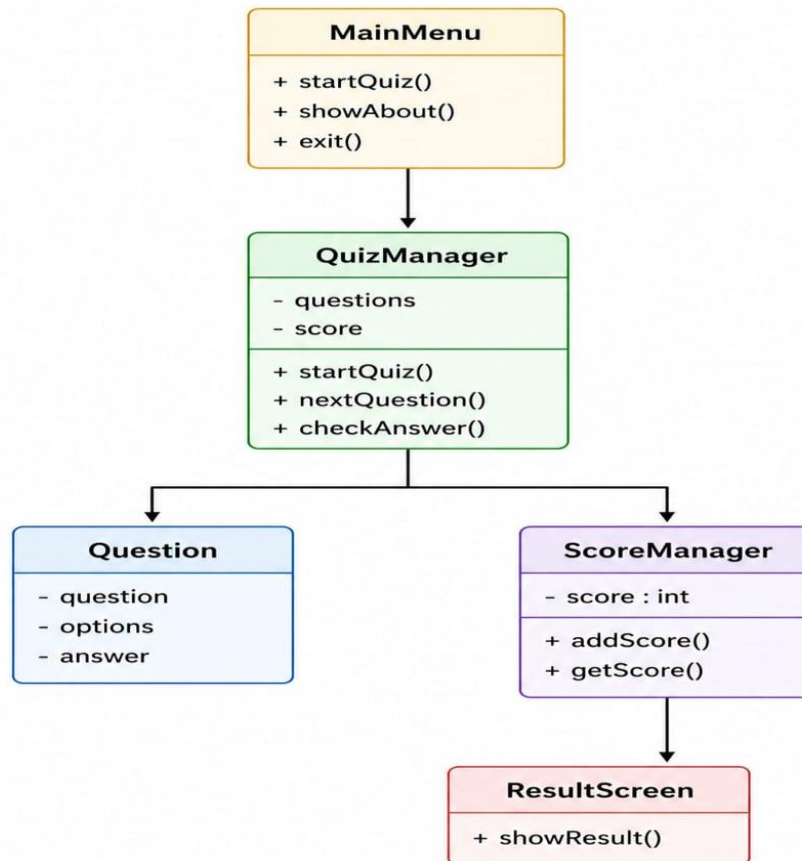
The goal of Week 2 was to design the full software architecture of the quiz game before any actual coding begins. This included:

- Designing the software architecture and deciding how the application will be structured.
- Creating a class diagram to map out all classes and their relationships.
- Designing the application workflow — how the user moves through the quiz from start to finish.
- Planning the user interface layout for each screen of the application.

2. Detailed Task Breakdown

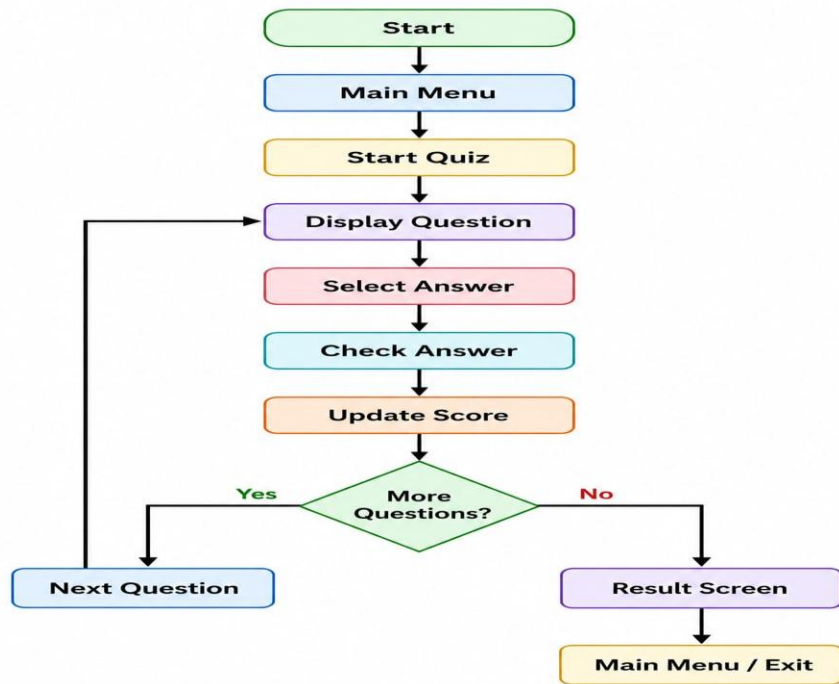
Software Architecture Design

Designed the overall structure of the quiz application using Object-Oriented Programming (OOP) principles. Decided how the application will be split into classes, what each class is responsible for, and how they communicate with one another. This planning step ensures the code stays clean, organised, and easy to extend throughout the rest of the project.



Class Diagram

Created a class diagram to visually represent all planned classes and their relationships. The diagram covers the core classes of the quiz game — such as the main application window, question management, answer evaluation, and the scoring system — and shows which classes depend on or interact with each other. This diagram acts as the blueprint for the full implementation phase starting from Week 3.



Application Workflow Design

Mapped out how the user moves through the application from launch to finish. The flow starts at the Main Menu where the user can choose Start Quiz, About, or Exit. When the quiz starts, questions are shown one at a time and the user selects an answer. The score is updated after each answer. Once all questions are completed, the application moves to a Results Screen showing the final score, with an option to return to the Main Menu or quit.

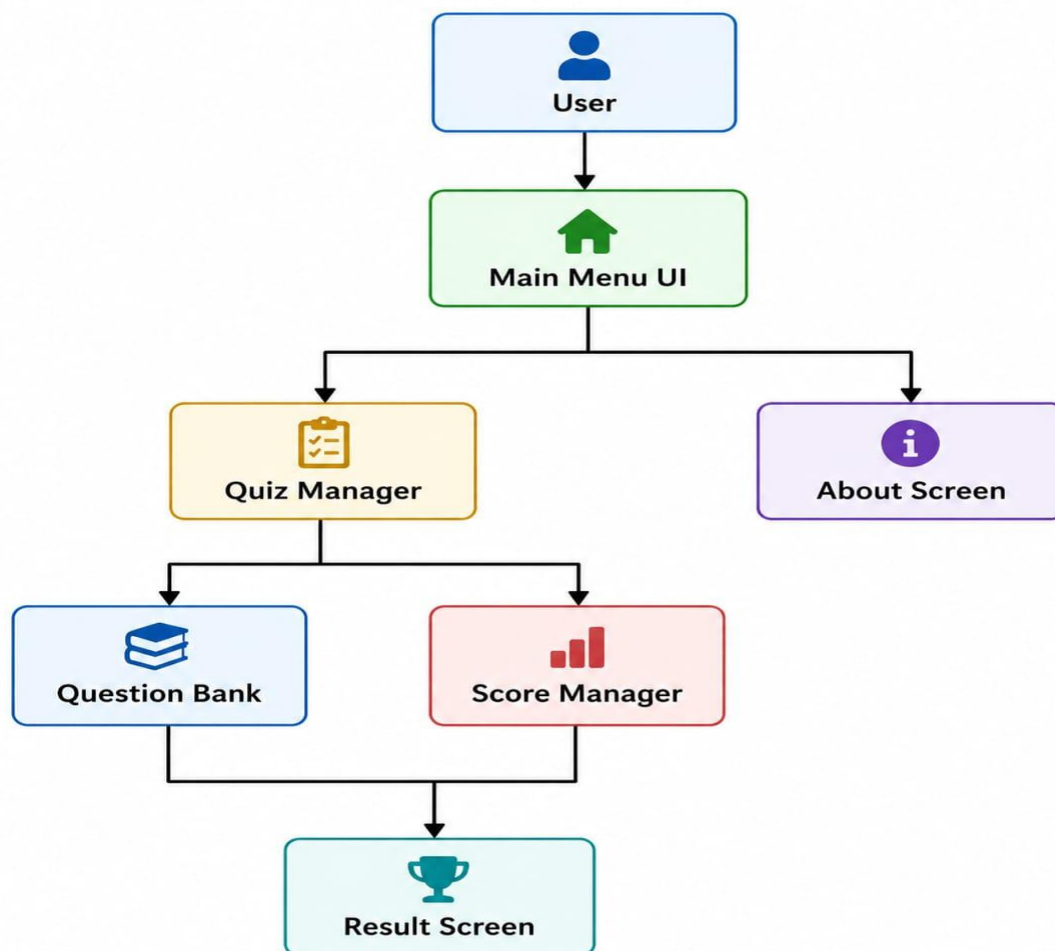


Figure 3.2: Application Workflow Diagram of Quiz App

User Interface Layout Planning

Planned the layout of each screen before building anything:

- Main Menu Screen: Application title at the top, three centred buttons below — Start Quiz, About, and Exit.
- Quiz Screen: Current question displayed at the top, four answer option buttons below it, and a progress/score indicator at the bottom showing the current question number out of total.
- Result Screen: Final score shown prominently in the centre, a short performance message, and a button to return to the Main Menu.
- About Screen: A brief description of the application with a Back button to return to the Main Menu.

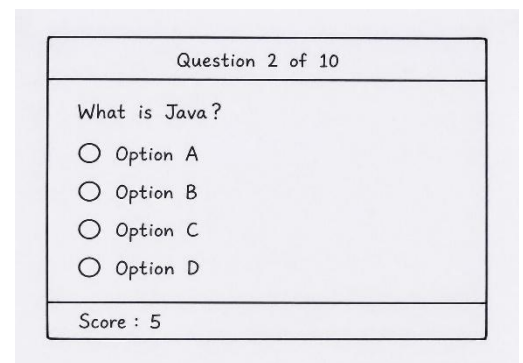
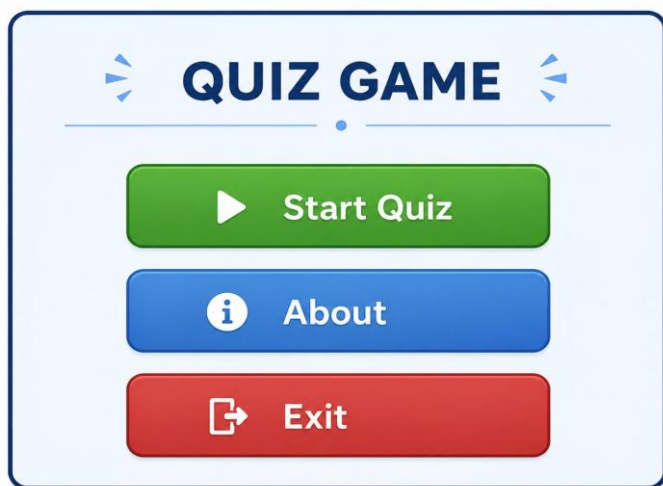


Figure 6: Result Screen Design



Figure 7: About Screen Design

3. What I Need to Learn

The following topics were studied and researched during Week 2:

- Object-Oriented Programming (OOP) — understanding classes, objects, inheritance, and encapsulation.
- UML Class Diagram — how to read and draw class diagrams to represent software structure.
- Java Classes and Objects — how to define and use classes and objects in Java.
- Software Design Principles — keeping the design simple, modular, and easy to extend.

4. Deliverables

- System Design Document: A written description of the software architecture and design decisions.
- Class Diagram: A diagram showing all planned classes and their relationships.
- Project Structure: A planned folder and file structure for the project codebase.

5. What I Will Demonstrate

At the end of Week 2, the following will be presented:

- Software Architecture — an explanation of how the application is structured and the reasoning behind it.
- Workflow Design — a walkthrough of how the user moves through the application from start to finish.
- Class Diagram — the complete class diagram showing all planned classes and their relationships.

6. GitHub Upload

The following items will be uploaded to the GitHub repository at the end of Week 2:

- System Design Document
- Class Diagram

- Project Structure

7. Tools & Resources

Development Environment

- Java JDK 8+
- IntelliJ IDEA / VS Code

Reference Materials

- Java OOP documentation and tutorials.
- UML class diagram references and diagram drawing tools.
- Software design principles and best practices guides.